

REVIEW

GPUMD 4.0: A high-performance molecular dynamics package for versatile materials simulations with machine-learned potentials

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Abstract

This paper provides a comprehensive overview of the latest stable release of the graphics processing units molecular dynamics (GPUMD) package, GPUMD 4.0. We begin with a brief review of its development history, starting from the initial version. We then discuss the theoretical foundations for the development of the GPUMD package, including the formulations of the interatomic force, virial and heat current for many-body potentials, the development of the highly efficient and flexible neuroevolution potential (NEP) method, the supported integrators and related operations, the various physical properties that can be calculated on the fly, and the GPUMD ecosystem. After presenting these functionalities, we review a range of applications enabled by GPUMD, particularly in combination with the NEP approach. Finally, we outline possible future development directions for GPUMD.

KEYWORDS

GPUMD, interatomic potential, machine-learned potential, materials simulation, molecular dynamics

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1 | INTRODUCTION

The molecular dynamics (MD) simulation method is one of the most powerful atomistic simulation methods used to study material properties, ranging from the atomic to the micro and even the mesoscale. An MD package serves as the computational engine behind atomistic simulations, making it an essential tool for researchers in this field. Open-source MD packages play a pivotal role in the development of algorithms and their practical applications. Among the most widely used free and open-source MD packages are GRO-MACS,^[1] LAMMPS,^[2] and OpenMM,^[3] to name a few. The graphics processing units molecular dynamics (GPUMD) package, which also belongs to this group, is the subject of this review. While not yet as widely adopted as the aforementioned packages, GPUMD has been gaining popularity at a rapid pace (Figure 1). It has been included in the list maintained by Talirz et al.,^[4,5] which tracks trends and statistics in atomistic simulation engines, and exhibits the highest relative growth rate for the last 2 years. Since its first release in August 2017 (version 1.0)^[6] and the update in May 2022 (version 3.3.1),^[7] many new features have been added, warranting a comprehensive review.

GPUMD has many distinguishing features making it appealing to both users and developers. It is an MD package developed for heterogeneous CPU-GPU computing platforms from the ground up, like HOOMD-blue^[8] and GALAMOST^[9] (later updated to PyGAMD^[10]). It is also one of the first MD packages that incorporates native machine-learned potentials, which makes it applicable to numerous complex materials that are inaccessible to traditional empirical potentials. The machine-learned potentials in GPUMD can deliver near-quantum-mechanical accuracy at the speed of empirical potentials, enabling predictive and

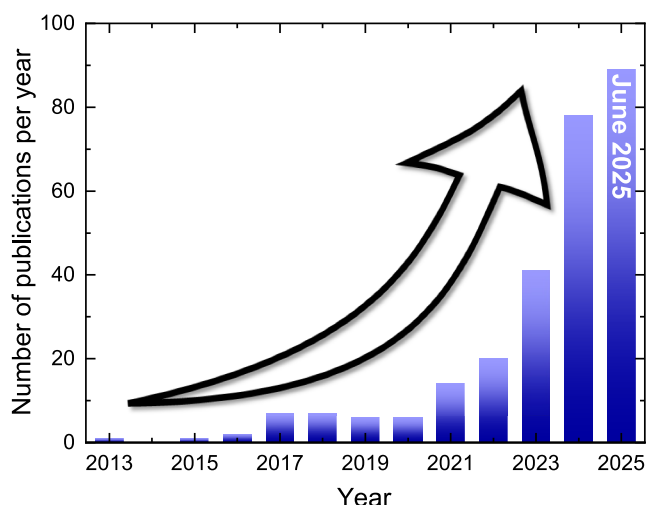


FIGURE 1 Number of publications (including preprints) per year using graphics processing units molecular dynamics, up to June 14, 2025.

efficient simulations of a wide range of processes and properties. In this paper, we give a comprehensive review and discussion of the past, present, and future of GPUMD.

2 | THE DEVELOPMENT HISTORY OF GPUMD

Although the first version of GPUMD was released in 2017,^[6] its development dates back to 2011 when it began as an exercise for a CUDA programming course. At that time, the package only supported the Lennard-Jones potential and its sole functionality was to calculate the thermal conductivity via the Green–Kubo method. This functionality was further developed in 2013, with improved computational efficiency for the Coulomb-Buckingham potential.^[11]

In 2015, a general formulation of force, virial, and heat current for many-body potentials was developed, providing the foundation for an efficient implementation of the heat current.^[12] This advancement led to an efficient GPU implementation^[6] of many-body potentials such as the embedded atom method,^[13,14] Stillinger–Weber,^[15] and Tersoff potentials.^[16] With these developments, the first version of GPUMD^[6] was released as an open-source software in 2017, containing about 10,000 lines of source code, written in CUDA C.

The next major development in GPUMD was the addition of the homogeneous nonequilibrium MD method^[17] and related spectral decomposition techniques^[17–19] during 2018 and 2019. These developments made GPUMD a popular package for heat transport applications.

In 2019, the development of interatomic potentials, such as a variant of the Tersoff potential^[20] and the so-called force-constant potential,^[21] began. However, the focus quickly shifted to general-purpose machine-learned potentials. In 2021, a native machine-learned potential, the neuroevolution potential (NEP),^[22] was developed. The NEP approach underwent several improvements^[17,23,24] from 2021 to 2024. The rapid growth in the popularity of GPUMD in recent years has been driven to a large extent by the development of the NEP approach, which provides highly efficient and accurate potential models for a wide range of materials.^[25]

The latest version of GPUMD, released^[26] in April 2025, is GPUMD 4.0, which we will describe here. It contains about 85,000 lines of source code written in CUDA C++. For simplicity, we will use GPUMD to refer to GPUMD 4.0 unless otherwise stated.

3 | CURRENT FEATURES IN GPUMD

We categorize the functionalities of GPUMD into three major areas: potentials, integrators, and properties. For a detailed discussion on CUDA programming aspects and the

physical foundations underlying GPUMD, we refer the interested reader to relevant textbooks.^[27,28] Before examining the three functional categories, we provide a concise overview of GPUMD, focusing on its practical usage.

GPUMD is primarily developed using CUDA C++ (although it has also been adapted to work with HIP). Upon compilation, two executables are generated: `gpumd` and `nep`. The `nep` executable serves the training of NEP models, whereas the `gpumd` executable is designed for conducting MD simulations. For the `nep` executable, two files are required:

1. `nep.in`: This file governs the training process.
2. `train.xyz`: This file contains the training data.

1. `run.in`: This file controls the MD simulation.
2. `model.xyz`: This file defines the system to be simulated.

Similarly, for the `gpumd` executable, at least two files must be provided:

Both the `train.xyz` and `model.xyz` files adhere to the standard extended XYZ file format. The `nep.in` file includes straightforward commands that specify the hyperparameters for NEP training.

In contrast, the `run.in` file is comparatively more complex and flexible. In the simplest cases, users only need to define the interatomic potential using the potential keyword and create `ensemble-run` blocks to specify the MD simulation process. Within an `ensemble-run` block, users can incorporate operations to modify the simulation process or compute and output useful quantities. More details will be discussed later, and comprehensive documentation is available at <https://gpumd.org/>.

3.1 | Interatomic potentials

Interatomic potentials describe the interactions between atoms and are required inputs to MD simulations. GPUMD supports both conventional empirical potentials and machine-learned potentials, as listed in Table 1.

3.1.1 | Empirical potentials

With respect to empirical potentials, GPUMD supports the 12-6 Lennard-Jones potential,^[29] the embedded atom method potential,^[13,14] the Tersoff potential,^[16] and the registry-dependent interlayer potential.^[33–36] The interlayer potential accurately describes anisotropic interlayer van der Waals interactions of layered materials and is usually used in combination with another potential for the intralayer interactions. Note that the NEP approach has been specifically implemented^[31] as an intralayer potential that retains the computational efficiency of traditional empirical potentials such as Tersoff while achieving near ab initio accuracy.

3.1.2 | Machine-learned potentials

For machine-learned potentials, GPUMD currently supports three types: the force-constant potential,^[21] the NEP, and the deep potential.^[30] Both force constant and deep potential models need to be trained using external packages, specifically the `hivest`^[37] and `DeePMD-kit`^[30] packages, respectively. The NEP approach, on the other hand, is a native machine-learned potential fully implemented in GPUMD, including both training and inference.

3.1.3 | Formulation of force, virial, and heat current

For all the interatomic potentials in GPUMD, the implementation follows the formalism established for general many-body potentials.^[12] All the potential models are defined in terms of the site energy U_i for a given atom i , whose summation gives the total potential energy of the system:

$$U = \sum_i U_i.$$

The site energy generally depends on its local environment and can be formally expressed as

TABLE 1 Interatomic potentials implemented in the graphics processing units molecular dynamics package.

Interatomic potential	Reference	Comments
Lennard-Jones (LJ)	[29]	The classical two-body potential
Embedded-atom method (EAM)	[13, 14]	Empirical many-body potential for metals
Tersoff	[16]	Empirical many-body potential for covalent bonds
Force-constant potential (FCP)	[21]	Machine-learned potential for equilibrium dynamics
Neuroevolution potential (NEP)	[7, 22–24]	General-purpose machine-learned potential
Deep potential (DP)	[30]	General-purpose machine-learned potential
Hybrid anisotropic interlayer potential (ILP) and NEP	[31]	For various Van der Waals structures
Hybrid ILP and Stillinger–Weber (SW) potential	[32]	For transition metal dichalcogenide structures

$$U_i = U_i(\{\mathbf{r}_{ij}\}_{j \in \mathcal{N}_i}),$$

where $\{\mathbf{r}_{ij}\}_{j \in \mathcal{N}_i}$ is the set of position differences from atom i to neighboring atoms $j \in \mathcal{N}_i$:

$$\mathbf{r}_{ij} \equiv \mathbf{r}_j - \mathbf{r}_i.$$

The force acting on atom i can be derived as follows:

$$\begin{aligned} \mathbf{F}_i &= -\frac{\partial}{\partial \mathbf{r}_i} \sum_j U_j \\ &= -\frac{\partial U_i}{\partial \mathbf{r}_i} - \frac{\partial}{\partial \mathbf{r}_i} \sum_{j \neq i} U_j \\ &= -\sum_{j \neq i} \frac{\partial U_i}{\partial \mathbf{r}_{ij}} \frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{r}_i} - \sum_{j \neq i} \sum_{k \neq j} \frac{\partial U_j}{\partial \mathbf{r}_{jk}} \frac{\partial \mathbf{r}_{jk}}{\partial \mathbf{r}_i} \\ &= \sum_{j \neq i} \frac{\partial U_i}{\partial \mathbf{r}_{ij}} - \sum_{j \neq i} \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \\ &= \sum_{j \neq i} \left(\frac{\partial U_i}{\partial \mathbf{r}_{ij}} - \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \right). \end{aligned}$$

This establishes the validity of the (weak form of) Newton's third law. That is, for a general many-body potential, there exists a pair-wise force

$$\mathbf{F}_{ij} = \frac{\partial U_i}{\partial \mathbf{r}_{ij}} - \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \quad (1)$$

between any pair of atoms i and j that fulfills

$$\mathbf{F}_{ij} = -\mathbf{F}_{ji}.$$

After realizing the existence of the above pairwise force, the virial tensor and heat current can be elegantly formulated. Starting from the definition of the virial tensor, we have

$$\begin{aligned} \mathbf{W} &\equiv \sum_i \mathbf{r}_i \otimes \mathbf{F}_i \\ &= \sum_i \sum_{j \neq i} \mathbf{r}_i \otimes \mathbf{F}_{ij} \\ &= \sum_i \sum_{j \neq i} \mathbf{r}_i \otimes \left(\frac{\partial U_i}{\partial \mathbf{r}_{ij}} - \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \right) \\ &= \sum_i \sum_{j \neq i} \mathbf{r}_i \otimes \frac{\partial U_i}{\partial \mathbf{r}_{ij}} - \sum_i \sum_{j \neq i} \mathbf{r}_i \otimes \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \\ &= \sum_j \sum_{i \neq j} \mathbf{r}_j \otimes \frac{\partial U_j}{\partial \mathbf{r}_{ji}} - \sum_i \sum_{j \neq i} \mathbf{r}_i \otimes \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \\ &= \sum_i \sum_{j \neq i} \mathbf{r}_{ij} \otimes \frac{\partial U_j}{\partial \mathbf{r}_{ji}}. \end{aligned} \quad (2)$$

There are a few equivalent expressions for the virial tensor. For example, it can also be expressed as

$$\mathbf{W} = -\sum_i \sum_{j \neq i} \mathbf{r}_{ij} \otimes \frac{\partial U_i}{\partial \mathbf{r}_{ij}}.$$

However, the expression in the last line of Equation (2) is more convenient in heat transport applications. To see this, we derive the heat current from its definition:

$$\mathbf{J} \equiv \frac{d}{dt} \sum_i \mathbf{r}_i \left(U_i + \frac{1}{2} m_i \mathbf{v}_i^2 \right) = \mathbf{J}^{\text{pot}} + \mathbf{J}^{\text{kin}},$$

where $\mathbf{J}^{\text{kin}} = \sum_i \mathbf{v}_i (U_i + \frac{1}{2} m_i \mathbf{v}_i^2)$ is the kinetic part of the heat current. The potential part can be further derived as follows:

$$\begin{aligned} \mathbf{J}^{\text{pot}} &= \sum_i \mathbf{r}_i \frac{d}{dt} \left(U_i + \frac{1}{2} m_i \mathbf{v}_i^2 \right) \\ &= \sum_i \mathbf{r}_i \left[\sum_{j \neq i} \frac{\partial U_i}{\partial \mathbf{r}_{ij}} \cdot (\mathbf{v}_j - \mathbf{v}_i) + \mathbf{F}_i \cdot \mathbf{v}_i \right] \\ &= \sum_i \mathbf{r}_i \sum_{j \neq i} \left[\frac{\partial U_i}{\partial \mathbf{r}_{ij}} \cdot (\mathbf{v}_j - \mathbf{v}_i) + \left(\frac{\partial U_i}{\partial \mathbf{r}_{ij}} - \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \right) \cdot \mathbf{v}_i \right] \\ &= \sum_i \mathbf{r}_i \sum_{j \neq i} \left[\frac{\partial U_i}{\partial \mathbf{r}_{ij}} \cdot \mathbf{v}_j - \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \cdot \mathbf{v}_i \right] \\ &= -\frac{1}{2} \sum_i \sum_{j \neq i} \mathbf{r}_{ij} \left[\frac{\partial U_i}{\partial \mathbf{r}_{ij}} \cdot \mathbf{v}_j - \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \cdot \mathbf{v}_i \right] \\ &= -\sum_i \sum_{j \neq i} \mathbf{r}_{ij} \frac{\partial U_i}{\partial \mathbf{r}_{ij}} \cdot \mathbf{v}_j \\ &= \sum_i \sum_{j \neq i} \mathbf{r}_{ij} \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \cdot \mathbf{v}_i. \end{aligned} \quad (3)$$

The last three lines in Equation (3) are all legitimate expressions of the heat current in periodic systems. The last line is, however, a more convenient one in practical implementation, as it only involves the velocity $\mathbf{v}_i$ of the central atom i , and not the velocities $\mathbf{v}_j$ of the neighboring atoms j . Based on this consideration, we define a per-atom virial according to Equation (2):

$$\mathbf{W}_i = \sum_{j \neq i} \mathbf{r}_{ij} \otimes \frac{\partial U_j}{\partial \mathbf{r}_{ji}} \quad (4)$$

such that $\mathbf{W} = \sum_i \mathbf{W}_i$ and

$$\mathbf{J}^{\text{pot}} = \sum_i \mathbf{W}_i \cdot \mathbf{v}_i. \quad (5)$$

Therefore, the per-atom virial expression in Equation (4) is the basis for both pressure and heat current calculations in GPUMD.

From Equations (1) and (4), it is evident that the terms $\partial U_i / \partial \mathbf{r}_{ij}$ and $\partial U_j / \partial \mathbf{r}_{ji}$ are crucial in these calculations. The term $\partial U_i / \partial \mathbf{r}_{ij}$ is known as the partial force,^[12] and the other term can be obtained by exchanging indices ($i \leftrightarrow j$). Thus, the calculations of force, virial (pressure), and heat current in GPUMD ultimately hinge on the calculation of partial forces. This elegant formulation is fundamental for the efficient GPU implementation of many-body potentials without resorting to atomic functions.^[6]

It is worth emphasizing that the formulation above applies to all potential models in GPUMD. Given that NEP is the most commonly used potential model in GPUMD, we discuss its formulation in more detail below.

3.1.4 | Neuroevolution potentials

The NEP approach generally follows the Behler–Parinello neural network potential methodology,^[38] but it differs in terms of the atomic-environment descriptor and the training method. Specifically, we describe the latest version of NEP here, known as NEP4.^[24]

In NEP4, the site energy U_i for a given atom i is a function of an abstract descriptor vector $\mathbf{q}^i$ with a number of components q_v^i ($v = 1, 2, \dots, N_{\text{des}}$). Each descriptor component characterizes the structural and chemical environments of atom i partially. The descriptor components are divided into two groups, one with radial dependence only, called radial descriptors, and the other with additional angular dependence, called angular descriptors.

The radial descriptors are labeled by the index n and are constructed as a sum of radial functions over the neighboring atoms:

$$q_n^i = \sum_{j \neq i} g_n(r_{ij}). \quad (6)$$

The radial function $g_n(r_{ij})$ is constructed as a linear combination of a set of $N_{\text{bas}}^R + 1$ basis functions:

$$g_n(r_{ij}) = \sum_{k=0}^{N_{\text{bas}}^R} c_{nk}^{JJ} f_k(r_{ij}). \quad (7)$$

The basis functions $f_k(r_{ij})$ are defined as

$$f_k(r_{ij}) = \frac{1}{2} \left[T_k \left(2(r_{ij}/r_c^R - 1)^2 - 1 \right) + 1 \right] f_c(r_{ij}),$$

where $T_k(x)$ is the k -th order Chebyshev polynomial of the first kind. The function $f_c(r_{ij})$ is a smoothing function defined as

$$f_c(r_{ij}) = \begin{cases} \frac{1}{2} [1 + \cos(\pi r_{ij}/r_c^R)], & r_{ij} \leq r_c^R; \\ 0, & r_{ij} > r_c^R, \end{cases}$$

where r_c^R is a cutoff radius beyond which the basis functions are zero. The chemical species are embedded in the expansion coefficients c_{nk}^{IJ} of the radial functions, where I and J indicate the types of atoms i and j . These coefficients are trainable, resulting in different radial functions for different pairs of atoms.

The angular descriptors depend both on the radial distances r_{ij} , and the angles θ_{ijk} formed by the $\mathbf{r}_{ij}$ and $\mathbf{r}_{ik}$ vectors,

$$\cos \theta_{ijk} = \frac{\mathbf{r}_{ij} \cdot \mathbf{r}_{ik}}{r_{ij} r_{ik}}.$$

The simplest angular descriptors in NEP are defined in terms of the Legendre polynomials $P_l(x)$:

$$q_{nl}^i = \frac{2l+1}{4\pi} \sum_{j \neq i} \sum_{k \neq i} g_n(r_{ij}) g_n(r_{ik}) P_l(\cos \theta_{ijk}). \quad (8)$$

The radial and angular dependencies are indicated by the subscripts n and l in q_{nl}^i . Note that the radial functions in q_{nl}^i are defined similarly to Equation (7) but a different cutoff radius r_c^A and expansion order N_{bas}^A can be used. Efficient evaluation of the angular descriptors requires transforming the Legendre polynomial to spherical harmonics. There are also other types of angular descriptors in NEP. For more details, we refer to ref. [7].

The descriptor vector $\mathbf{q}^i$ is assembled from the radial and angular descriptors described above. Then, the site energy in NEP is formally written as $U_i(\mathbf{q}^i)$. Currently, only a single hidden layer is used in the neural network model for NEP, and the site energy can be explicitly written as

$$U_i = \sum_{\mu=1}^{N_{\text{neu}}} w_{\mu}^{(1)} \tanh \left(\sum_{v=1}^{N_{\text{des}}} w_{\mu v}^{(0)} q_v^i - b_{\mu}^{(0)} \right) - b^{(1)}. \quad (9)$$

Here, $\tanh(x)$ is the activation function, $w^{(0)}$ are the weight parameters connecting the input layer (with dimension N_{des}) and the hidden layer (with dimension N_{neu}), $w^{(1)}$ represents the weight parameters connecting the hidden layer and the output layer (the site energy), $b^{(0)}$ represents the bias parameters in the hidden layer, and $b^{(1)}$ is the bias parameter in the output layer. All these parameters are trainable, similar to the expansion coefficients in the radial functions.

In terms of the descriptor vector, the partial force can be written as

$$\frac{\partial U_i}{\partial \mathbf{r}_{ij}} = \sum_{v=1}^{N_{\text{des}}} \frac{\partial U_i}{\partial \mathbf{q}_v^i} \frac{\partial \mathbf{q}_v^i}{\partial \mathbf{r}_{ij}}$$

With the partial force available, force, virial, and heat current can all be readily evaluated. The derivative $\partial \mathbf{q}_v^i / \partial \mathbf{r}_{ij}$ could be evaluated using auto-differentiation techniques. However, we opted to derive explicit and simplified expressions by hand and implemented them using native CUDA kernels. This approach reduces external dependencies of the GPUMD package and optimizes its computational performance.

NEP can be used in combination with other potentials.^[39,40] In addition to the interlayer potential mentioned above, it can also be used in combination with the DFT-D3 potential^[41] and the Ziegler-Biersack-Littmark potential.^[42] The DFT-D3 potential can capture weak van der Waals interactions, whereas the Ziegler-Biersack-Littmark potential is usually used to ensure the physicality of the (repulsive) interaction when atoms get very close to each other.

Recently, Liang et al.^[43] developed NEP89, a comprehensive foundation model covering virtually the entire periodic table, which has been released alongside GPUMD 4.0. NEP89 can be used out of the box or as a starting point that can be conveniently fine-tuned with a relatively small amount of additional training data for system-specific applications. NEP89 achieves competitive accuracy compared to representative foundation models across a range of properties, including energetics, elastic properties, phonon frequencies, and MD simulations of amorphous carbon, liquid water, and solid-state electrolytes, while being three to four orders of magnitude more computationally efficient, enabling large-scale atomistic simulations of inorganic and organic systems that were previously impractical. The NEP89 model is the first one in a series of NEP-based foundation models, which are expected to exhibit better capabilities in future versions.

The NEP approach, as implemented in GPUMD, achieves exceptionally high computational performance. Using a single GPU, either professional or consumer grade, NEP-GPUMD can achieve a computational speed of the order of 10^7 atom-step per second in typical MD simulations. The number of atoms in the simulation cell manageable with a single GPU ranges from 1 to 10 million, depending on the device memory of the GPU. With eight 80-GB A100 GPUs, NEP has been used to simulate a high-entropy alloy with 100 million atoms.^[24] The outstanding computational performance advantage of NEP over other representative GPU-based machine-learned potentials, such as deep potential (DP),^[30] NequIP,^[44] and MACE,^[45] is clearly demonstrated in Table 2.

3.2 | Integrators and related operations

3.2.1 | The velocity-Verlet integrator

Integrators solve the equations of motion and are central to the atomic dynamics in MD simulations. Without any external control, an isolated system adheres to Hamiltonian dynamics,

TABLE 2 Computational speed (in units of atom-step per second per GPU) of neuroevolution potential (NEP) (within graphics processing units molecular dynamics) compared to representative GPU-based machine-learned potentials (MLPs), including deep potential (DP),^[30] NequIP,^[44] and MACE,^[45] all interfaced to LAMMPS.^[2]

Material	Speed	MLP	GPU	Reference
2D Silicene	870,000	DP	V100	[22]
	13,000,000	NEP	V100	[22]
Bulk PbTe	520,000	DP	V100	[22]
	11,000,000	NEP	V100	[22]
Tungsten	130,000	DP	A100	[39]
	13,000,000	NEP	A100	[39]
Diamond	1200	MACE	V100	[25]
	2000	NequIP	V100	[25]
	1,830,000	NEP	V100	[25]

resulting in a microcanonical ensemble. In this ensemble, the number of particles N , the volume V (or more precisely the simulation cell), and the total energy E of the system remain constant. Hence, it is known as the NVE ensemble. The velocity-Verlet integrator^[46] is used in GPUMD, which can be formulated more formally in terms of the Liouville operator and the Trotter decomposition.^[47] The integration for one time step Δt can be expressed as follows:

$$\begin{aligned} \mathbf{r}_i(t + \Delta t) &\approx \mathbf{r}_i(t) + \mathbf{v}_i(t)\Delta t + \frac{1}{2} \frac{\mathbf{F}_i(t)}{m_i} (\Delta t)^2; \\ \mathbf{v}_i(t + \Delta t) &\approx \mathbf{v}_i(t) + \Delta t \frac{\mathbf{F}_i(t) + \mathbf{F}_i(t + \Delta t)}{2m_i}. \end{aligned}$$

In GPUMD, a statistical ensemble is specified by the `ensemble` keyword, followed by a specific ensemble type. For example, the NVE ensemble is invoked by the combined keyword `ensemble nve`. The keywords for the various integrators/ensembles and related operations are listed in Table 3.

3.2.2 | Thermostats and barostats

Other statistical ensembles can be realized by controlling temperature T and/or pressure P . By controlling the temperature only, we have the NVT ensemble. Here, the energy is not constant, but the temperature has a well-defined mean value in equilibrium. GPUMD supports several thermostats for temperature control, including the Berendsen thermostat^[48] (`ensemble nvt_ber`), the Nosé-Hoover chain thermostat^[47] (`ensemble nvt_nhc`), the Bussi-Donadio-Parrinello thermostat (also called stochastic velocity rescaling thermostat)^[49] (`ensemble nvt_bdp`), and the Langevin thermostats^[50,51] (`ensemble nvt_lan` and `ensemble nvt_bao`). By controlling the pressure as well,

TABLE 3 Integrators and related operations implemented in the graphics processing units molecular dynamics package.

Integrators/ensembles	Keyword
<i>NVE</i>	ensemble nve
Berendsen <i>NVT</i>	ensemble nvt_ber
NHC <i>NVT</i>	ensemble nvt_nhc
BDP <i>NVT</i>	ensemble nvt_bdp
Langevin <i>NVT</i>	ensemble nvt_lan
Langevin <i>NVT</i>	ensemble nvt_bao
Berendsen <i>NPT</i>	ensemble npt_ber
SCR <i>NPT</i>	ensemble npt_scr
MTTK <i>NPT</i>	ensemble npt_mttk
MTTK-based <i>NPH</i>	ensemble nph_mttk
NEMD heat transport	ensemble heat_nhc
NEMD heat transport	ensemble heat_bdp
NEMD heat transport	ensemble heat_lan
Equilibrium TI	ensemble ti
Nonequilibrium TI	ensemble ti_spring
Nonequilibrium TI, RS path	ensemble ti_rs
Nonequilibrium TI, AS path	ensemble ti_as
Nonequilibrium TI, liquid	ensemble ti_liquid
Hugoniotat shock method	ensemble nphug
NEMD shock piston	ensemble wall_piston
NEMD shock mirror	ensemble wall_mirror
NEMD shock harmonic	ensemble wall_harmonic
MSST	ensemble msst
PIMD	ensemble pimd
RPMD	ensemble rpmd
TRPMD	ensemble trpmd
Canonical MC	mc canonical
SGC-MC	mc sgc
VCSGC-MC	mc vcsgc
Change box once	change_box
Deform box during a run	deform
Fix a group of atoms	fix
Move a group of atoms	move
Add external forces to atoms	add_force
Add electric field to ions	add_efield
Add stopping forces to atoms	electron_stop

Abbreviations: AS, adiabatic switching; BDP, Bussi-Donadio-Parrinello; MSST, multi-scale shock technique; MTTK, Martyna-Tuckerman-Tobias-Klein; NEMD, nonequilibrium molecular dynamics; NHC, Nose-Hoover chain; PIMD, path-integral molecular dynamics; RPMD, ring-polymer molecular dynamics; RS, reversible scaling; SCR, stochastic cell rescaling; SGC, semi-grand canonical; TI, thermodynamic integration; TRPMD, thermostatted RPMD; VCSGC, variance-constrained SGC.

we have the *NPT* ensemble. GPUMD supports a few barostats for pressure control, including the Berendsen barostat^[48] (ensemble npt_ber), the Martyna-Tuckerman-Tobias-Klein barostat^[47] (ensemble npt_mttk), and the Bernetti-Bussi barostat (also called stochastic cell rescaling barostat)^[52] (ensemble nvt_scr). If pressure is controlled but temperature is not, we have the *NPH* ensemble which conserves the enthalpy H in equilibrium. This has only been implemented in the Martyna-Tuckerman-Tobias-Klein approach^[47] (ensemble nph_mttk).

The thermostats can also be applied locally to a group of atoms to enable nonequilibrium MD simulations for applications such as heat transport. These include the Bussi-Donadio-Parrinello thermostat (ensemble heat_bdp), the Nose-Hoover chain thermostat (ensemble heat_nhc), and the Langevin thermostat (ensemble heat_lan). Among these, the Langevin thermostat is recommended for heat transport applications.^[53]

3.2.3 | Thermodynamic integration and free energy calculations

We have implemented a series of thermodynamic integration methods for Helmholtz and Gibbs free-energy calculations, which are useful for studying phase diagrams. These include the equilibrium approach (ensemble ti) and nonequilibrium approach (ensemble ti_spring). These methods only apply to solids, and the free energies are calculated in reference to the Einstein crystal.^[54] The nonequilibrium approaches allow for efficient integration along a reversible scaling path^[55] (ensemble ti_rs) or an adiabatic switching path^[56] (ensemble ti_as). For liquids (ensemble ti_liquid), the free energies are calculated in reference to the Uhlenbeck-Ford model.^[57]

3.2.4 | Shock simulation methods

We have implemented a few shock methods.^[58] One method is based on the constant stress Hugoniotat method^[59] (ensemble nphug). With a target stress, this algorithm adjusts the temperature to make the system converge to the Hugoniot. Another method is based on the multiscale shock technique^[60] (ensemble msst). Besides, there are a few nonequilibrium MD methods, where a shock wave is generated by a moving wall, which can be a fixed layer of atoms (piston) (ensemble wall_piston), a momentum mirror that reflects atoms (ensemble wall_mirror), or a harmonic potential that pushes atoms away (ensemble wall_harmonic).

3.2.5 | Path-integral methods

The above integrators are for classical MD simulations. Nuclear quantum effects can be partially captured by path-

integral MD simulation methods, which have recently been implemented into GPUMD.^[61] Apart from the normal path-integral MD algorithm based on the Langevin thermostat and normal modes^[60] (ensemble `pimd`), we also implemented the ring-polymer MD^[62] (ensemble `rpmd`) and thermostatted ring-polymer MD^[63] (ensemble `trpmd`). For the normal modes, we used the robust integration algorithm based on the Cayley transform.^[64]

3.2.6 | Hybrid Monte Carlo and molecular dynamics

MD simulations can be supplemented by Monte Carlo simulations to enable sampling of the compositional degrees of freedom in mixed (alloyed) systems. Recently, a series of hybrid Monte Carlo and MD simulation methods has been implemented into GPUMD,^[65] including the canonical Monte Carlo ensemble (exchanging pairs of atoms of different species) (`mc canonical`), the semi-grand canonical Monte Carlo ensemble (flipping the species of single atoms) (`mc sgc`), and the variance-constrained semi-grand canonical Monte Carlo ensemble^[66,67] (`mc vcs gc`). The choice of input parameters in terms of normalization follows the expressions in ref. [68].

3.2.7 | Other operations

Apart from the various integrators/ensembles, the time evolution can also be altered by other related operations. The `change_box` keyword can be used to deform the box instantly, and the `deform` keyword can be used to deform the box during a run. A group of atoms can be fixed (frozen) by using the `fix` keyword or be moved as a rigid body using the `move` keyword. External forces can be added to the atoms via the `add_force` keyword, and external electric forces can be added to ions (charged atoms) via the `add_efield` keyword. Frictional forces on fast-moving atoms due to electronic collisions can be invoked by the `electron_stop` keyword, which is useful in irradiation damage or ion implantation simulations.

3.3 | Properties

The usefulness of a MD package is finally manifested in the physical properties that can be calculated using it. GPUMD supports dumping of trajectories and moreover enables the calculation of many useful quantities on the fly (Table 4).

3.3.1 | The dump-like keywords

During any MD simulation, it is recommended to dump the basic thermodynamic quantities for the whole system, using

TABLE 4 Dump and compute keywords implemented in the graphics processing units molecular dynamics package.

Properties	Keyword
Thermodynamic quantities	<code>dump_thermo</code>
Trajectory and related quantities	<code>dump_xyz</code>
Trajectory for the beads in PIMD	<code>dump_beads</code>
Restarting file	<code>dump_restart</code>
Space-time average	<code>compute</code>
RDF	<code>compute_rdf</code>
ADF	<code>compute_adf</code>
Angular-dependent RDF	<code>compute_angular_rdf</code>
SDC	<code>compute_sdc</code>
MSD	<code>compute_msd</code>
Viscosity	<code>compute_viscosity</code>
EMD thermal transport	<code>compute_hac</code>
HNEMD thermal transport	<code>compute_hnemd</code>
HNEMDEC thermal transport	<code>compute_hnemdec</code>
Spectral decomposition	<code>compute_shc</code>
Modal decomposition (EMD)	<code>compute_gkma</code>
Modal decomposition (HNEMD)	<code>compute_hnema</code>
Electronic transport	<code>compute_lsqt</code>
Phonon properties	<code>compute_phonon</code>
VDOS	<code>compute_dos</code>

Abbreviations: ADF, angular distribution function; EMD, equilibrium molecular dynamics; HNEMD, homogeneous nonequilibrium molecular dynamics; HNEMDEC, HNEMD with Evans-Cummings algorithm; MSD, mean-square displacement; RDF, radial distribution function; RPMD, ring-polymer molecular dynamics; SDC, self-diffusion coefficient; VDOS, vibrational density of states.

the `dump_thermo` keyword. The quantities dumped include temperature, kinetic energy, potential energy, pressure tensor, and the simulation cell metric.

Trajectories and related quantities can be dumped using the `dump_xyz` keyword. This will generate an output file in the extended XYZ format that can be visualized by programs such as OVITO.^[69] In this file, the trajectory is stored frame by frame. Each frame contains $N + 2$ lines, where N is the number of atoms in the frame that is written in the first line. The second line contains information such as the global time, boundary conditions, cell metric, total energy, virial, and stress. The next N lines then contain the atom symbols, positions, and possibly other per-atom quantities. A related keyword for path-integral MD simulations is `dump_beads`. Finally, the `dump_restart` keyword can be used to save a file that can be used to restart a simulation, although the present implementation does not lead to a perfect restart in some cases, such as those involving the Nosé–Hoover chain thermostat, where some thermostating variables are not saved in the restart file. This is expected to be improved during the next code refactoring process.

3.3.2 | The compute-like keywords

The compute-like keywords are used to calculate physical quantities on the fly. The simplest keyword is `compute`, which calculates the spatial and time averages of various quantities. This is useful for, for example, getting a temperature profile or a stress distribution. There are also keywords for common structural properties, such as the radial distribution function (`compute_rdf`), the angular distribution function (`compute_adf`), and the angular-dependent radial distribution function (`compute_ardf`).

The majority of the compute-like keywords are related to transport properties. These usually involve time-correlation functions, making on-the-fly calculations valuable for minimizing data storage. The transport properties include the self-diffusion coefficient from the velocity autocorrelation function (`compute_sdc`) or the mean-square displacement (`compute_msd`), the viscosity (`compute_viscosity`), the thermal conductivity from heat current autocorrelation function (`compute_hac`), the thermal conductivity from homogeneous nonequilibrium MD simulations^[17] (`compute_hnemd` and `compute_hnemdec`), the spectral (`compute_shc`)^[17] and modal^[79] (`compute_hnema` and `compute_gkma`) decompositions as well as electronic transport properties from the linear-scaling quantum transport methods^[70] (`compute_lsqt`). The linear-scaling quantum transport calculations are based on tight-binding models.

GPUMD also supports the direct calculation of phonon properties, such as phonon dispersions using the finite-displacement method (`compute_phonon`) and vibrational density of states from the mass-weighted velocity autocorrelation function (`compute_vdos`).

4 | THE GPUMD ECOSYSTEM

Although the `nep` and `gpumd` executables are standalone programs that operate without external dependencies, they can be complemented by other tools and packages, collectively forming the GPUMD ecosystem.

4.1 | Tools within the GPUMD package

The tools are included with the `tools` directory of the GPUMD package. Most utilities in the `tools` directory focus on the preparation and analysis of training and test datasets for NEP. For example, `abacus2xyz`, `castep2xyz`, `cp2k2xyz`, `orca2xyz`, and `vasp2xyz` are designed to convert outputs from various quantum chemistry packages into training/test datasets formatted in the extended XYZ format. Similarly, `dp2xyz`, `mtp2xyz`, and `runner2xyz` facilitate the conversion of training datasets from other formats into the extended XYZ format.

4.2 | Additional related packages

Additional related packages are listed in Table 5, most of which are based on the `NEP_CPU` package. This package contains a standalone C++ implementation of the inference of NEP, which serves as the computational engine for many Python-based packages listed in Table 5. Furthermore, it provides an interface to the LAMMPS package,^[2] enabling NEP to function in more computing environments.

The `calorine` package^[71] is a versatile Python library designed to construct and use NEP models, offering ASE (Atomic Simulation Environment)^[79] calculators, input/output functions for GPUMD files, NEP model inspection, descriptor space analysis, structure generation, and NEP training, and can be easily used to perform various calculations, including relaxation, phonon properties, elastic properties, free energy calculations, thermal conductivity via the Boltzmann transport equation, and so on.

The `dynasor` package^[72,73] is designed for calculating total and partial dynamic structure factors as well as related correlation functions from MD simulations, including particularly those driven by GPUMD. By analyzing these functions, one can access the dynamics of a system without resorting to perturbative approaches. It can also predict

TABLE 5 Packages and repositories related to graphics processing units molecular dynamics (GPUMD) and/or neuroevolution potential (NEP).

Packages	Code repository
NEP_CPU	https://github.com/brucefan1983/NEP_CPU
calorine ^[71]	https://gitlab.com/materials-modeling/calorine
dynasor ^[72,73]	https://gitlab.com/materials-modeling/dynasor
GPUMD-Wizard	https://github.com/Jonsnow-willow/GPUMD-Wizard
Gpyumd	https://github.com/AlexGabourie/gpyumd
GPUMDkit	https://github.com/zhyan0603/GPUMDkit
MAGUS ^[74,75]	https://gitlab.com/bigd4/magus
mdapy ^[76]	https://github.com/mushroomfire/mdapy
NepTrain ^[77]	https://github.com/aboys-cb/NepTrain
NepTrainKit ^[77]	https://github.com/aboys-cb/NepTrainKit
NEP_Active	https://github.com/psn417/NEP_Active
nep_maker	https://github.com/psn417/nep_maker
PyNEP	https://github.com/bigd4/PyNEP
PySED ^[78]	https://github.com/Tingliangstu/pySED
Somd	https://github.com/initqp/somd
nep-data	https://gitlab.com/brucefan1983/nep-data
GPUMD-Tutorials	https://github.com/brucefan1983/GPUMD-Tutorials

experimental spectra by combining the structure factor with the cross sections or form factors.

GPUMD-Wizard, a material structure processing software based on ASE, automates the calculation of various material properties, including lattice constants, elastic constants, and defect formation energies, while also facilitating the execution and analysis of MD simulations using GPUMD.

The `gpyumd` package is a collection of tools that generate valid input files and process the output files of GPUMD. It leverages the functionality of ASE when beneficial, but is otherwise independent to remain flexible and best serve GPUMD directly.

The MAGUS package^[74,75] is a machine learning and graph theory assisted crystal structure prediction package. It has interfaces for various quantum chemistry packages and machine-learned potentials, including NEP.

GPUMDkit, a shell-based toolkit for the GPUMD and NEP, offers a user-friendly command-line interface to streamline common scripts and workflows, simplifying tasks such as script invocation, format conversion, structure sampling, NEP construction workflow, and various analyses, aiming to improve user productivity.

The `mdapy`^[76] Python library provides an array of powerful, flexible, and straightforward tools to analyze atomic trajectories generated from MD simulations. It well supports the extended XYZ format output by GPUMD and takes advantage of the highly parallel processing capabilities on multicore CPUs and GPUs to provide excellent efficiency and flexibility for processing and analyzing trajectories.

NepTrain and NepTrainKit are Python packages designed to enhance the construction of NEP models, with NepTrain integrating tools for active learning workflows, including structural perturbations, configurational space exploration, single-point energy calculations, and force field training, whereas NepTrainKit provides user-friendly visualization and processing of NEP training datasets, enabling detailed analysis of dataset composition and model performance.

NEP_Active and `nep_maker` also focus on NEP model construction, with NEP_Active employing active learning strategies to automate training set construction and `nep_maker` extending this by incorporating a comprehensive workflow to automate active learning by submitting and monitoring jobs.

PyNEP serves as a Python interface for NEP, providing ASE calculators, descriptor calculations for atoms, and phonon calculations, but is particularly noted for its implementation of the farthest point sampling method to select representative structures.

PySED^[78] is a Python-based package built upon the spectral energy density method, designed to analyze specific phonon-mode information from large-scale MD trajectories, enabling convenient calculation of kinetic-energy-weighted phonon dispersions and derivation of phonon lifetimes. It was developed to work with NEP-driven MD simulations.

The `somd` package includes a simple wrapper for the `nep` executable, enabling automatic construction of NEP models through active learning strategies.

Finally, we note that numerous training and test datasets related to NEP have been compiled in the `nep-data` repository, although the collection is not exhaustive. Additionally, the GPUMD-Tutorials repository offers a wide range of valuable tutorials and examples, covering various practical aspects of the GPUMD package.

5 | APPLICATIONS OF GPUMD TO MATERIALS CALCULATIONS

To date, GPUMD has been utilized in approximately two hundred publications. Table 6 provides a comprehensive list of these publications, highlighting the first authors and the primary materials investigated.

5.1 | Applications in early years

Prior to 2022, GPUMD applications focused predominantly on covalently bonded systems, which have traditionally been described using the Tersoff potential. Consequently, the range of materials studied during this period was quite limited, primarily comprising two-dimensional materials such as graphene, hexagonal BN, and MoS₂. Thermal transport was the main theme of these early investigations.

Building on these materials, significant advancements were made in computational methods for heat transport. These include the unambiguous definition of heat current for general many-body potentials,^[12] the demonstrated equivalence between equilibrium and nonequilibrium MD methods,^[87] the development of a general formulation for the homogeneous nonequilibrium MD method along with related spectral decomposition techniques,^[17] the examination of impact of thermostating methods on the nonequilibrium MD method,^[53] and the interpretation of apparent thermal conductivity using the equilibrium MD method.^[106]

Beyond the methodological progress, GPUMD has also been employed to uncover the physical mechanisms underlying phonon thermal transport in various materials. A particularly noteworthy application involved studying heat transport in multilayer MoS₂, successfully reproducing the experimentally observed highly anisotropic thermal transport.^[108]

5.2 | Applications with NEP

Since 2022, GPUMD has been employed to study a wider range of materials, thanks to the development of the NEP approach in 2021^[22] and its improvements in 2022^[7,23] and 2024.^[24] Heat transport remained a major application area

TABLE 6 Applications of the graphics processing units molecular dynamics package in various materials.

Year	Publications (major materials)
2013	Fan ^[11] (Ar, PbTe)
2015	Fan ^[12] (Si, C)
2016	Hirvonen ^[80] (C); Mortazavi ^[81] (C)
2017	Azizi ^[82] (C); Fan ^[6] (C); Fan ^[18] (C); Fan ^[83] (C); Fan ^[84] (C); Hirvonen ^[85] (C); Mortazavi ^[86] (C)
2018	Dong ^[87] (Si, C); Dong ^[88] (BN); Fan ^[89] (C); Hirvonen ^[90] (C); Mortazavi ^[91] (CN); Rajabpour ^[92] (C); Xu ^[93] (P)
2019	Fan ^[17] (C, Si); Fan ^[20] (Si); Gu ^[94] (C); Isaeva ^[95] (Si); Li ^[53] (C); Xu ^[96] (MoS ₂)
2020	Bea ^[97] (Si); Dong ^[98] (C); Fu ^[99] (Si); Gabourie ^[100] (MoS ₂); Wu ^[101] (C); Wu ^[102] (BN)
2021	Barbalinardo ^[103] (C); Chen ^[104] (CFs); Dong ^[105] (C/BN); Dong ^[106] (Si, C); Du ^[107] (C/BN); Fan ^[22] (PbTe, Si); Gabourie ^[19] (HfO ₂ , SiO ₂); Kim ^[108] (MoS ₂); Lundgren ^[109] (SiGe); So ^[110] (C); Wang ^[111] (C); Wu ^[112] (C/BN); Wu ^[113] (C/BN); Zhang ^[114] (C)
2022	Cheng ^[115] (Si-Ge); Dong ^[116] (Si); Fan ^[23] (PbTe); Fan ^[7] (PbTe, C); Feng ^[117] (C); Gabourie ^[118] (MoS ₂); Jin ^[119] (Si, Ge); Li ^[120] (Al-Mg); Li ^[121] (Si); Li ^[122] (Si); Liang ^[123] (C); Sha ^[124] (C/BN); Sha ^[125] (CN); Wang ^[126] (Li ₆ Al); Wu ^[127] (C, BN); Wu ^[128] (C); Wu ^[129] (C); Xu ^[130] (C); Ying ^[131] (C); Zhou ^[132] (Si)
2023	Bea ^[133] (Si); Cheng ^[134] (PbTe); Cheng ^[135] (C); DeVries ^[136] (MX ₂ (M = Mo, W; X = S, Se)); Dong ^[137] (C); Du ^[138] (PH ₄ AlBr ₄); Eriksson ^[139] (C, BN, MoS ₂); Fransson ^[140] (CsPbBr ₃ , MAPbI ₃); Fransson ^[141] (CsPbBr ₃); Fransson ^[142] (CsPbX ₃ (X = Cl, Br, I)); Li ^[143] (C); Liang ^[144] (SiO ₂); Liu ^[39] (W); Liu ^[145] (Si/Ge); Lu ^[146] (C); Lu ^[147] (C); Ouyang ^[148] (AgX (X = Cl, Br, I)); Pan ^[149] (MgOH); Rosander ^[150] (BaZrO ₃); Sha ^[151] (PbTe); Shi ^[152] (C); Shi ^[153] (InGeX ₃ (X = S, Se, Te)); Shi ^[154] (CsPbX ₃ (X = Cl, Br, I)); Su ^[155] (Cs ₂ BiAgBr ₆ , Cs ₂ BiAgCl ₆); Sun ^[156] (Ga ₂ O ₃); Wang ^[157] (Si); Wang ^[158] (SrTiO ₃); Wei ^[159] (C); Wiktor ^[160] (CsMX ₃ (M = Sn, Pb and X = Cl, Br, I)); Wu ^[161] (C, BN); Wu ^[162] (C, C ₃ N); Xiong ^[163] (C); Xu ^[164] (H ₂ O); Yang ^[165] (GaN/C); Ying ^[166] (C); Ying ^[40] (MOF); Ying ^[167] (P); Ying ^[168] (MOF); Zhang ^[169] (HfO ₂); Zhao ^[170] (Pd-Cu-Ni-P); Zhou ^[171] (Ge-Si, Ge)
2024	Berger ^[172] (MoS ₂ , BAs); Berger ^[173] (Amino acids); Berrens ^[174] (H ₂ O); Cao ^[175] (PC); Chen ^[176] (C, BN); Chen ^[177] (GeSn); Chen ^[178] (H ₂ O); Cheng ^[179] (A ₂ SnBr ₆ (A=Rb, Cs)); Cheng ^[180] (SiGe); Deng ^[181] (Si); Dong ^[182] (Si); Dong ^[183] (ScAlN); Fan ^[184] (MOF); Fan ^[70] (C); Fang ^[185] (CH); Feng ^[186] (C); Fine ^[187] (Ca ₃ CrN ₃ H); Folkner ^[188] (Si); Fransson ^[189] (MAPbI ₃); Fransson ^[190] (BaZrO ₃); Gabourie ^[191] (Si, SiO ₂ , HfO ₂); Huang ^[192] (C); Huang ^[193] (Mg ₃ (Sb, Bi) ₂); Li ^[194] (Gr); Li ^[195] (C); Li ^[196] (Sb-Te); Li ^[197] (Si); Li ^[198] (C ₉ H ₄ BO ₂); Li ^[199] (C); Liu ^[200] (HECs); Lyu ^[201] (PbSe); Muhammed ^[202] (Perovskites); Oliveira ^[203] (Si); Pan ^[58] (SiO ₂); Pegolo ^[204] (Li _x Si _{1-x}); Qi ^[205] (AlN, C); Ru ^[206] (PdSe ₂); Schäfer ^[207] (PTA); Shi ^[208] (BN); So ^[209] (Ga ₂ O ₃ , BN); So ^[210] (C); Song ^[24] (16 metals); Sonti ^[211] (Zeolite-Confined Gold); Sun ^[212] (Co, Mo, Fe, Ni, Cu); Sun ^[213] (AlN, C); Sun ^[214] (Ga ₂ O ₃ /C); Tang ^[215] (BN); Tang ^[216] (ScF ₃); Tian ^[217] (NaCl-CaCl ₂); Tian ^[218] (H ₂ O); Timalsina ^[219] (MgNiCoCuZnO ₅); Wan ^[220] (C ₆ N ₇); Wang ^[221] (COFs); Wang ^[222] (H ₂ O); Wang ^[223] (Si); Wang ^[224] (Ga ₂ O ₃); Wei ^[225] (high-entropy rare-earth monosilicates); Wu ^[226] (C); Wu ^[227] (C); Wu ^[228] (Si, GaAs, C, PbTe); Wu ^[229] (C); Wu ^[230] (C, BN); Xu ^[231] (Perovskite); Xu ^[232] (Perovskite); Yan ^[233] (Li ₇ La ₃ Zr ₂ O ₁₂); Yang ^[234] (C); Ying ^[235] (C); Yu ^[236] (C); Yu ^[237] (BN); Yue ^[238] (Si/C); Zeraati ^[239] (La ₂ Zr ₂ O ₇); Zhang ^[240] (BiI ₃); Zhang ^[241] (C); Zhang ^[242] (GeTe); Zhang ^[243] (Alanine dipeptide and acetyl chloride); Zhang ^[244] (Li-Be); Zhao ^[245] (Ti-Al-Nb); Zhou ^[246] (LiH)
2025	Ariana ^[247] (Na ₃ SbSe ₄); Bao ^[248] (LiNbO ₃ /LN); Berger ^[73] (Ni ₃ Al, BaZrO ₃); Bro-Jørgensen ^[249] (Au); Bu ^[31] (C/MoS ₂ /BN); Cao ^[250] (LiTFSI/G ₃); Cao ^[251] (Alloys); Chen ^[252] (Si:H); Chen ^[253] (Al-Cu-Li); Chen ^[254] (GaN); Chen ^[77] (CsPbI ₃); Cheng ^[255] (AgTl ₂ I ₃); Dai ^[256] (C aerogels); Donadio ^[257] (C); Du ^[258] (Sn ₄ Se ₁₀); Feng ^[259] (SiO ₂); Hainer ^[260] (MA _{1-x} FA _x PbI ₃); Hao ^[261] (CuBiSeCl ₂); Hu ^[262] (CL-20); Hu ^[263] (MoS ₂ /WSe ₂); Jia ^[264] (Zr); Jiang ^[36] (C/BN); Jiang ^[265] (MoS ₂); Jiang ^[32] (MX ₂ (M = Mo, W; X = S, Se)); Jiang ^[266] (MX ₂ (M = Mo, W; X = S, Se, Te)); Kayastha ^[267] (BaZrS ₃); Laven ^[268] (CsPbI ₃); Li ^[269] (Ga ₂ O ₃); Li ^[270] (BeGeN ₂); Li ^[271] (KTa _{1-x} Nb _x O ₃); Li ^[269] (COF); Li ^[272] (Al, Cu, Ag); Liang ^[273] (C/BN); Linderalv ^[274] (4H-SiC); Lindgren ^[275] (Si, C ₆ H ₆ , BaTi _{1-x} Se _x O ₃ H _x); Liu ^[276] (BN); Liu ^[277] (Ti); Liu ^[278] (Mg); Liu ^[279] (Cu ₇ PS ₆); Liu ^[280] (HEDs); Liu ^[281] (AlN-SiC); Lu ^[282] (C); Lu ^[283] (Aerogels); Luo ^[284] (N-Ga-Al); Lyu ^[285] (Pb-Se-Te-S); Moon ^[286] (C); Oh ^[287] (Si); Ouyang ^[288] (argyrodite); Pegolo ^[289] (Li ₃ PS ₄); Rosander ^[290] (BaZrO ₃); Seifi ^[291] (GaAs@InAs); Sun ^[292] (GaN/C); Sun ^[293] (NbSe ₃); Sun ^[294] (HfO ₂); Tan ^[295]

(Continues)

TABLE 6 (Continued)

Year	Publications (major materials)
	(C/BN); Tuchinda ^[296] (Alloys); Wang ^[297] (C); Wang ^[298] ((AlAs) _n /(InAs) _n); Wang ^[299] (Ne); Wang ^[300] (C/BN); Wang ^[301] (BaTiS ₃); Wang ^[302] (GeTe/Sb ₂ Te ₃); Wang ^[303] (β -Ga ₂ O ₃); Wen ^[304] ((Hf, Ta, Zr, W)C); Wu ^[305] (MoSe ₂ /WSe ₂); Xiao ^[306] (C); Xiao ^[307] (AgSnSbTe ₃); Xu ^[308] (LiF); Xu ^[309] (tobermorite); Yan ^[310] (Li ₇ La ₃ Zr ₂ O ₁₂); Yang ^[311] (Si); Yuan ^[312] (MgO, LiH); Yue ^[313] (Si/Ge); Zeng ^[314] (Cs ₃ Bi ₂ I ₆ Cl ₃); Zeraati ^[315] (TBCCOs); Zhang ^[316] (MoSi ₂ N ₄); Zhang ^[317] (SiC); Zhang ^[318] (C); Zhang ^[319] (Penta-PdSe ₂); Zhang ^[320] (C/polydimethylsiloxane); Zhang ^[321] (Al ₂ O ₃); Zhang ^[322] (W-La); Zhang ^[323] (C); Zhou ^[324] (C); Zhou ^[325] (NbTaZr); Zhou ^[326] (BAs); Zhou ^[327] (Ga ₂ O ₃ /BAs); Zhou ^[328] (In ₂ Se ₃); Zhu ^[329] (HEC/Cr ₇ C ₃)

Note: The table includes publications (including preprints) up to June 14, 2025.

for GPUMD, as reviewed by Dong et al.^[182] up to March 2024. Nevertheless, numerous other application fields have also emerged, as summarized by Ying et al.^[25] up to January 2025. Below, we briefly outline the various research fields, emphasizing key publications that pioneered the use of GPUMD in combination of NEP in these areas.

5.2.1 | Mechanical properties

Ying et al.^[166] were the first to apply GPUMD and NEP to investigate mechanical properties in their study of a C₆₀-based quasi-two-dimensional network. Their work demonstrated consistent results for quasi-static deformation processes when compared with quantum-mechanical calculations. Moreover, they extended the scope of their study to larger spatial and temporal scales, approaching strain rates that are almost experimentally attainable.

5.2.2 | Radiation damage

Liu et al.^[39] expanded the NEP approach by incorporating the Ziegler-Biersack-Littmark potential,^[42] applying it for the first time to investigate primary radiation damage in tungsten. They conducted large-scale MD simulations involving up to 8.1 million atoms over 240 ps using a single 40-GB A100 GPU, achieving computational efficiency comparable to that of embedded-atom-method potentials. Their study also highlighted the superior accuracy of the NEP model over embedded-atom-method potentials in capturing radiation damage in foils.

5.2.3 | Phase transition

The accuracy and efficiency offered by NEP models have also facilitated in-depth studies of phase transitions. Fransson et al.^[142] were the first to investigate temperature-induced structural phase transitions in inorganic halide perovskites using GPUMD and NEP. Their work revealed the impact of simulation size, temperature variation rate, and the choice of exchange-correlation functionals in quantum-mechanical calculations for training data.

5.2.4 | Shock simulation

Shi et al.^[152] initiated the study of shock compression using GPUMD and NEP. They developed a NEP model for carbon at high pressures, which demonstrated exceptional capabilities in modeling both the melting behavior and the Hugoniot line. They designed a thermodynamic pathway suitable for double shock compression experiments, facilitating the discovery of the long-sought BC8 phase of carbon.

5.2.5 | Short-range order

Chen et al.^[177] initiated the study of short-range order in GeSn alloys using GPUMD and NEP. A compact yet representative dataset was constructed via farthest-point sampling to improve training efficiency and predictive accuracy. Through extensive statistical sampling, they uncovered intricate short-range order features that strongly impact the electronic band structure. Large-scale simulations revealed the coexistence of nanoscale short-range order domains, which is promising for optoelectronic applications.

5.2.6 | Ion transport

Yan et al.^[233] initiated research on ion transport in solid-state electrolytes, a critical area for advancing all-solid-state battery technology. They developed a NEP model to explore the effects of lithium nonstoichiometry on ionic conductivity and phase stability in Li₇La₃Zr₂O₁₂. Their findings revealed that even minor deviations from stoichiometry, particularly lithium deficiency, significantly lower the activation energy for Li⁺ diffusion in tetragonal Li₇La₃Zr₂O₁₂. This leads to a remarkable ten-orders-of-magnitude increase in room-temperature ionic conductivity.

5.2.7 | Electronic transport

Electronic and transport properties can be studied effectively using linear-scaling quantum transport approaches.^[330] Fan et al.^[70] combined these techniques with MD simulations,

showcasing their feasibility in modeling the electronic and thermoelectric transport properties of complex materials at finite temperatures.

5.2.8 | Tensorial properties

The NEP approach has also been successfully extended to modeling tensorial properties, such as electric dipole moments and polarizability, by Xu et al.^[232] They demonstrated the effectiveness of this method in predicting infrared and Raman spectra for various systems, including liquid water, single molecules, and a prototypical perovskite exhibiting strong anharmonicity.

5.2.9 | Large NEP models

Song et al.^[24] pioneered the development of large NEP models for multiple species, emphasizing the importance of focusing on elemental and binary systems during data construction. They successfully created a general-purpose NEP model for 16 metals and their arbitrary alloys, demonstrating substantially higher accuracy compared to the conventional embedded-atom method. This NEP model also achieved a computational milestone by simulating 100 million atoms using only eight 80-GB A100 GPUs.

5.2.10 | Hybrid Monte Carlo and molecular dynamics

Song et al.^[65] were the first to apply NEP in conjunction with hybrid Monte Carlo and MD simulations. The Monte Carlo sampling with NEP is highly efficient, as it leverages the locality of the potential function. This approach holds significant promise for investigating the effects of chemical order in multicomponent systems.

5.2.11 | Nuclear quantum effects

Path-integral MD simulations are essential for accurately capturing nuclear quantum effects in materials. A highly efficient GPU implementation of these simulations has recently been integrated into GPUMD. Ying et al.^[61] demonstrated the effectiveness of this approach by investigating thermal properties for several materials, including lithium hydride, porous metal–organic frameworks, liquid water as well as elemental aluminum.

5.2.12 | Melting in confined systems

Wang et al.^[299] have recently used GPUMD and NEP to study the melting transition in atomistically confined layered

materials. They developed NEP models for noble gases and aluminum confined between two graphene sheets at different pressures and temperatures. Although noble gases and aluminum typically form only close-packed structures, even under the extreme conditions of white dwarf stars, they discovered tetragonal-packed configurations in the confined systems. Upon heating, they found that confined two-dimensional monolayers melt according to the two-step continuous Kosterlitz-Thouless-Halperin-Nelson-Young theory. However, multilayer solids transition continuously into an intermediate layered-hexatic phase before melting discontinuously into an isotropic liquid. This change could be qualitatively explained based on a crossover from two-dimensional topological defects to three-dimensional ones during melting as the number of layers increases.

5.2.13 | Hybrid NEP and anisotropic interlayer potential

Bu et al.^[31] developed a hybrid computational framework that integrates a machine-learned potential, based on the NEP formalism, for intralayer interactions, with physics-based registry-dependent interlayer potential that captures anisotropic van der Waals interactions. This framework achieves near ab initio accuracy with a computational efficiency at the level of empirical potentials, enabling large-scale MD simulations of twisted van der Waals heterostructures.

5.2.14 | Chemical reactions

Deciphering chemical reaction mechanisms at the atomic scale is fundamental to advancing materials science, catalysis, and energy technologies. Liu et al.^[278] constructed a reactive NEP model for magnesium corrosion studies by utilizing high-temperature MD to sample critical interfacial regions within substrate-interface-solution systems. This enabled the atomistic simulation of the MgOH intermediate phase formation and revealed the inward oxide/hydroxide migration mechanism governing corrosion layer growth. Hu et al.^[262] also constructed a reactive NEP model to predict the reaction kinetics of 2,4,6,8,10,12-Hexanitro-2,4,6,8,10,12-hexaazaisowurtzitan (CL-20), a high energy-density material, under extreme conditions, revealing its atomic-level shock compression and thermal decomposition mechanisms. These studies demonstrate NEP's capability to resolve complex reaction pathways.

6 | SUMMARY AND FUTURE DIRECTIONS

In summary, we have provided a comprehensive overview of the GPUMD package, covering its development history, theoretical foundations, functionalities, and applications.

Although GPUMD is a relatively young MD package, it has been developing at a fast pace. Its robust theoretical foundation, based on an elegant formulation of many-body interatomic potentials, coupled with a well-designed GPU parallelism scheme and a versatile general-purpose machine-learned potential framework, has attracted increasing attention from researchers interested in exploring its capabilities.

Beyond its user base, the GPUMD package has also attracted numerous developers from around the globe. These contributors are working collaboratively to enhance its feature set, versatility, reliability, and efficiency, making the package increasingly robust and adaptable.

In the coming years, our efforts will focus on advancing GPUMD by further expanding its capabilities and enhancing its versatility. Building on the NEP approach, we will prioritize the incorporation of the charge degrees of freedom. This will pave the way for tackling a broader range of problems, such as those related to batteries and corrosion, further broadening the scope of applications for GPUMD. There are multiple possible strategies for combining NEP and the charge degree of freedom. The basic foundation is that the ion charges interact with each other via Coulomb forces, the evaluation of which can be either based on *k*-space techniques or a simple cutoff scheme. The NEP energy is then fitted to the difference between the target energy (usually based on quantum-mechanical calculations) and the Coulomb energy. The ion charge can be fitted to a particular type of target partial charge or simply be determined implicitly.^[331]

Additionally, we will work on building coarse-grained models based on the NEP approach. These models will significantly extend both the spatial and temporal scales achievable in MD simulations with GPUMD, opening up new possibilities for studying large systems and long-time phenomena. One straightforward strategy is force-matching, where the forces acting between the coarse-grained sites are systematically derived from the net forces experienced by those sites in all-atom simulations based on an all-atom NEP model. The coarse-grained system can also be described by a coarse-grained NEP model that is mathematically similar to the all-atom NEP model. Moreover, the separable natural evolution strategy in NEP is a powerful derivative-free black-box optimizer, which can conveniently take into consideration of experimental data as well as complex physical quantities during the training to enhance the reliability of the coarse-grained NEP models.

To push the boundaries of efficiency and accuracy, we plan to develop Monte Carlo sampling methods, enhanced sampling methods, and other time-acceleration techniques. These advancements will enable GPUMD to overcome the limitations of conventional MD simulations, allowing for a more comprehensive exploration of complex systems.

A direction that is seemingly unrelated to MD simulations is the development of general-purpose tight-binding models for electrons. These models will be an extension of the NEP framework, bridging the strengths of quantum transport methodologies^[330] and MD to enable more

accurate and efficient simulations of electronic properties in spatially complex materials.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The source code for GPUMD (version 4.0) is available at the Zenodo repository <https://doi.org/10.5281/zenodo.15299>

684^[26] and the GitHub repository <https://github.com/bruce-fan1983/GPUMD/releases/tag/v4.0>.

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